

Modeling Tennis Match Dynamics with Hidden Markov Models

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1. Motivation

Tennis matches evolve point-by-point, but underlying “momentum” or pressure states are not directly observable. This project uses statistical modeling to uncover hidden match dynamics and understand how match context evolves over time.

1. Data Overview

All data was collected or downloaded using Jeff Sackmann's **TennisAbstract** (<https://tennisabstract.com/>) project.

Dataset Summary **Point(s):** ~1.5 million points **Date Range:** 2000 to 2026 (*current*)

Key Variables

Feature	Meaning
first_serve_in_play	Whether first serve was successful
server_point_diff	Score difference
game_pressure	Pressure within the game
set_pressure	Pressure within the set
match_pressure	Overall match pressure
df_distance	Points since last double fault in current match

Note:
All features were standardized (mean = 0, std = 1) for comparability.

2. Relationships Between Variables

To evaluate whether the model inputs provide overlapping or independent information, we compute pairwise correlations between all features. This helps determine whether the feature set is appropriate for multivariate modeling (e.g., Hidden Markov Models).

Correlation Matrix

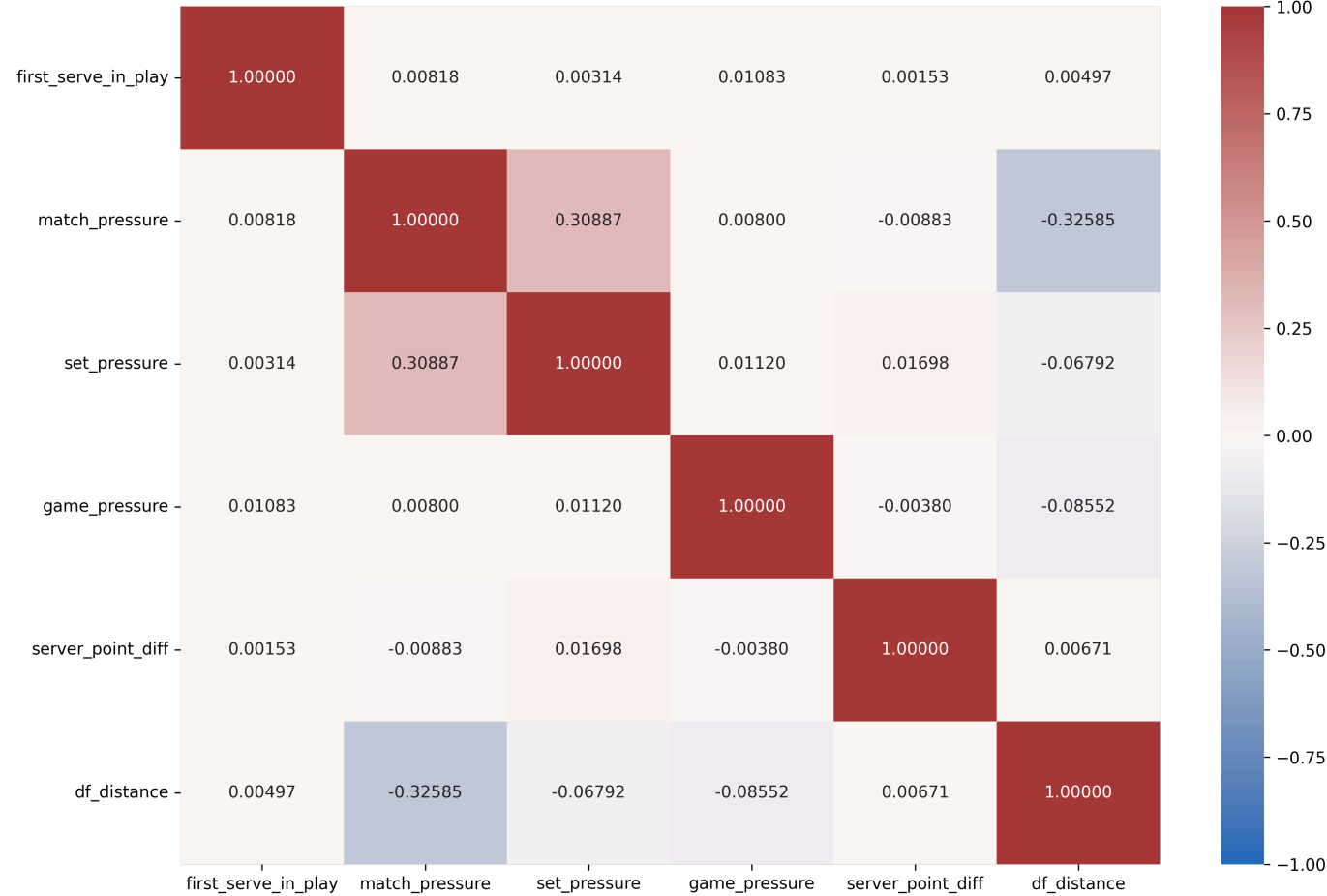
Feature	first_serve_in_play	match_pressure	set_pressure	game_pressure	server_point_diff	df_distance
first_serve_in_play	1.000000	0.008180	0.003141	0.010831	0.001531	0.004971
match_pressure	0.008180	1.000000	0.308872	0.008002	-0.008830	-0.325846
set_pressure	0.003141	0.308872	1.000000	0.011204	0.016980	-0.067924
game_pressure	0.010831	0.008002	0.011204	1.000000	-0.003804	-0.085523
server_point_diff	0.001531	-0.008830	0.016980	-0.003804	1.000000	0.006709
df_distance	0.004971	-0.325846	-0.067924	-0.085523	0.006709	1.000000

Interpretation

The low overall correlation suggests that each feature captures distinct aspects of match dynamics. This supports their use as inputs in a multivariate Hidden Markov Model.

Visualization

Feature Correlation Matrix



5. Temporal Behavior (Autocorrelation)

Feature	Lag 1	Lag 50	Interpretation
match_pressure	0.98	0.10	Very persistent
set_pressure	0.92	0.13	Strong carryover
df_distance	0.84	0.12	Smooth changes
game_pressure	0.51	~0.00	Moderate memory
server_point_diff	0.56	~0.00	Moderate memory
first_serve_in_play	0.04	~0.00	Nearly random

Takeaway:

- Pressure variables evolve slowly
- Serve success behaves almost randomly

6. Modeling Approach

Why Hidden Markov Models (HMM)?

- Observed variables are noisy
- Underlying match “states” are hidden
- Sequential structure matters

Model Used

- Gaussian Hidden Markov Model

- Handles continuous features
- Captures temporal dependencies

Inputs

- 6 standardized features per point

Structure

- Sequential modeling (point-by-point)
- Each match treated as its own sequence

7. Model Selection

Models evaluated with different numbers of hidden states.

Criteria:

- AIC (Akaike Information Criterion)
- BIC (Bayesian Information Criterion)

States	AIC	BIC
2	...	...
3	...	...
4	...	...
5	...	...

Selected Model:

Optimal number of states = **X**

8. Hidden State Interpretation

State	Characteristics
State 0	Low pressure, stable
State 1	High pressure
State 2	Transitional / volatile
State 3	Extreme conditions

(Replace with actual model outputs)

9. Transition Dynamics

- Transition matrix shows probabilities of moving between states

Key Observations:

- High diagonal values → states persist
- Transitions occur gradually

Interpretation:

Matches tend to remain in the same underlying condition for multiple points.

10. Baseline Comparison

Logistic Regression (Baseline)

- Uses same features
- No temporal modeling

Why Compare?

Demonstrates whether temporal modeling (HMM) improves performance.

Metrics to Include:

- Accuracy / AUC / Log-Likelihood
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11. Key Insights

- Match pressure evolves **slowly and predictably**
 - Serve success is **nearly random**
 - Hidden states capture **distinct match conditions**
 - Temporal modeling provides **additional structure**
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12. Limitations

- Missing or noisy data (serve outcomes)
 - Gaussian assumption for emissions
 - Hidden states may not map directly to real-world concepts
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13. Future Work

- Use **GMM-HMM** for more flexible distributions
 - Add player-specific features
 - Incorporate rally-level data
 - Predict match outcomes (win probability)
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14. Appendix (Optional)

Additional Details

- HMM components:
 - Transition matrix (A)
 - Emission distributions
 - Model assumptions
 - Additional plots:
 - ACF graphs
 - Feature distributions
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